Drone generation	Period/stage	Key challenges faced	Technological/strategic response
adaptive UAVs)		• Ethical and legal concerns • Data security and electronic warfare threats	• Advanced stealth design and sensor fusion • Blockchain and quantum encryption for secure control
6th generation (projected – cognitive UAVs)	Future	• Human–drone collaboration and trust • Complex swarm governance • Integration into smart city air networks	• Cognitive AI with human-machine collaboration • Quantum computing and neural network processing • Adaptive swarm intelligence frameworks